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Review

A Comparison of Data Quality Frameworks: A Review

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Abstract: This study reviews various data quality frameworks that have some form of regulatory backing. The aim is to identify how these frameworks define, measure, and apply data quality dimensions. This review identified generalisable frameworks, such as TDQM, ISO 8000, and ISO 25012, and specialised frameworks, such as IMF's DQAF, BCBS 239, WHO's DQA, and ALCOA+. A standardised data quality model was employed to map the dimensions of the data from each framework to a common vocabulary. This mapping enabled a gap analysis that highlights the presence or absence of specific data quality dimensions across the examined frameworks. The analysis revealed that core data quality dimensions such as "accuracy", "completeness", "consistency", and "timeliness" are equally and well represented across all frameworks. In contrast, dimensions such as "semantics" and "quantity" were found to be overlooked by most frameworks, despite their growing impact for data practitioners as tools such as knowledge graphs become more common. Frameworks tailored to specific domains were also found to include fewer overall data quality dimensions but contained dimensions that were absent from more general frameworks, highlighting the need for a standardised approach that incorporates both established and emerging data quality dimensions. This work condenses information on commonly used and regulation-backed data quality frameworks, allowing practitioners to develop tools and applications to apply these frameworks that are compliant with standards and regulations. The bibliometric analysis from this review emphasises the importance of adopting a comprehensive quality framework to enhance governance, ensure regulatory compliance, and improve decision-making processes in data-rich environments.

Keywords: data quality frameworks; data quality; data management; data regulations; data governance; TDQM; ISO 8000; ISO 25012; DAMA DMBoK; ALCOA+

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1. Introduction

Data quality (DQ) is defined by a set of values and attributes—often called dimensions—that can be qualitatively or quantitatively defined to describe the quality of datasets and other data structures [1,2]. The terminology used to describe these dimensions associated with data quality is complex, leading to standardisation efforts [2]. However, some commonly described dimensions include "accuracy", "completeness", "traceability", and "timeliness", which describe different aspects of a dataset. Given the increasing dependence on digital systems, primarily artificial intelligence (AI) and machine learning (ML) for informed decision making, having high-quality data is essential not only for ensuring operational efficiency but also to increase trust in these systems [3]. Standards, frameworks, guidelines, and regulations are commonly employed tools to ensure that high-quality data are used [1,4–13].

Data quality frameworks (DQFs) are structured methodologies used to assess, manage, and improve the quality of data. They can be directly built upon and supported by existing standards and regulations, or they can be specifically designed to address more tailored applications. These frameworks are essential for organisations to manage their data and demonstrate tangible evidence of the application of good data quality practices when communicating with stakeholders, both internal and external.

It is important to notice that this work is not the first attempt at providing an overview of existing DQFs, as there have been several attempts at doing so in recent years. Cichy et al. have published an overview of existing general-purpose data quality frameworks, where they offer a comprehensive systematic comparison between commonly known DQFs with the aim of informing data practitioners of the frameworks that better suit their needs [14]. While comprehensive, one limitation of this study that is recognised in their conclusions relates to the fact that several frameworks consider different dimensions of data quality, and no effort was done to map these using a common vocabulary or terminology. Additionally, the study also acknowledges the issue of regulatory compliance of the reviewed frameworks.

Similar studies of narrower scope have also been conducted, placing the focus on DQFs used only in specific domains such as healthcare [15,16] or finance [17]. While DQFs can be tailored for very specific applications, their wider applicability within each sector requires compliance with the established regulations of that sector. In many sectors, data management practices, and, by extension, data quality practices must be aligned with regulatory, governmental, or legislative standards [10–13]. This makes regulatory compliance a critical aspect of any DQF, ensuring that data management meets prespecified standards and increasing trust with stakeholders.

This paper provides a review of various DQFs that have received any type of regulatory backing across different sectors, offering a clear comparison of the sector-specific needs for high-quality data. Understanding how these regulated frameworks are used across different domains allows for the identification of common elements (i.e., DQ dimensions) between them. This compiled information can inform practitioners of the requirements of their specific domains, helping to avoid the mistakes associated with the use of unsuitable frameworks. This work builds upon previous work by the authors [2] that proposes a common terminology to describe data quality dimensions. This terminology is used to map dimensions between frameworks that have different nomenclatures, allowing a like-for-like comparison.

The significance of this review is its contribution to the understanding of DQFs and their regulatory compliance. It provides organisations and data practitioners with insights on how they can improve their data quality practices and ensure compliance with regulations. This information is equally relevant for emerging technologies, such as is the case with Large Language Model (LLM) AI systems [3]. This presents a rapidly evolving space that is cross-cutting over many different domains that regulators are struggling to keep up with. Leveraging existing regulated data quality frameworks provides a solid foundation for the development of AI-specific DQFs.

The frameworks covered in this paper include Total Data Quality Management (TDQM) [6]; Total Quality data Management (TQdM) [18]; ISO 8000 [1]; ISO 25012 [5]; Fair Information Practice Principles (FIPPS) [7]; the Quality Assurance Framework of the European Statistical System (ESS QAF) [8,19,20]; the IMF Data Quality Assessment Framework (DQAF) [10]; the UK Government Data Quality Framework [9,21]; the Data Management Body of Knowledge (DAMA DMBoK) [22,23]; the Basel Committee on Banking Supervision Standard (BCBS 239) [11]; the ALCOA+ Principles [24]; and the World Health Organization (WHO) Data Quality Assurance framework [13].

The remainder of this paper is structured as follows: Section 2 introduces general-purpose and foundational data quality frameworks; Section 3 presents data quality frameworks established as ISO standards; Section 4 discusses various governmental and international data quality frameworks; Section 5 highlights frameworks specifically used in the financial sector; Section 6 examines data quality frameworks employed in the healthcare sector; Section 7 collectively analyses and discusses all data quality frameworks reviewed by this work, comparing their common elements and identifying gaps in their assessments; and lastly, Section 8 summarises the findings of this review.

2. Data Management Frameworks

2.1. Total Data Quality Management (TDQM)

Total Data Quality Management (TDQM) is a holistic strategy for ensuring and monitoring the quality of data within organisations. TDQM was created in the 1980s by the MIT Sloan School of Management and was a pioneering research programme on data quality [6]. It has had a significant influence on the development of the data quality field [14,25–42]. It views data as a commodity and employs methodologies and strategies to ensure their high quality. It focuses on different dimensions of data quality that correspond to data quality categories. These are accuracy, objectivity, believability, reputation, access, security, relevance, value-added, timeliness, completeness, amount of data, interpretability, ease of understanding, concise representation, and consistent representation.

The application process for TDQM comprises four stages: definition, measurement, analysis, and improvement [25]. This is also know as the DMAI (Define, Measure, Analyse, Improve) cycle and is shown here in Figure 1. The definition phase involves determining the relevant dimensions of data quality for both the organisation and the specific data being considered. The measurement phase involves assessing the existing condition of data quality, identifying any issues, and understanding their effects on the organisation. The analysis phase includes investigating the fundamental reasons behind data quality challenges. Finally, the improvement phase involves executing modifications that address the identified challenges to enhance the quality of the data.

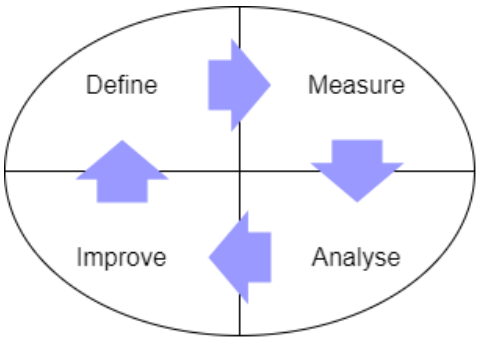


Figure 1. "Define, Measure, Analyse, Improve" cycle as outlined by TDQM [6,25] for refining adequate data management processes to implement.

This makes TDQM a method for executing the necessary cultural shift to establish a sustainable environment of ongoing data and information quality in an organisation. It offers a strong and all-encompassing framework for overseeing data quality. The framework acknowledges the critical importance of data in modern organisations and provides the necessary tools and methodologies to ensure the highest quality of this data.

2.2. Total Quality Data Management (TQdM)

Total Quality data Management (TQdM) shares the same holistic nature as TDQM. It consists of a comprehensive approach to managing data and information quality,

with added emphasis on improving quality through detailed process analysis, and assumes the role of a value system within an organisation that integrates data quality management beliefs, principles, and methods into the organisational culture [18]. Given its holistic nature, it has served as the basis for the development of multiple data quality assessment tools [30,41,43–52], including its reference in the ISO 8000 series.

TQdM can be broken down into six sequential and cyclical processes: "Assess data definition and information architecture quality", "Assess information quality", "Measure nonquality information costs", "Reengineer and cleanse data", "Improve information process quality", and "Establish the Information Quality Environment".

The first two processes consist of measurements and aim to create an initial understanding of the current quality infrastructure and metrics being employed. The third process uses these measurements to develop a value proposition for improving data quality. The fourth process tackles the technical aspects of improving the quality of the data being generated and used. The fifth process deals with integrating the fourth process into the existing data pipeline. Lastly, the sixth process condenses the learning and benefits yielded by previous processes to establish a new and upgraded data quality environment.

These processes are then repeated in a continuous cycle of improvement. This makes TQdM a similar framework for continuous data and information quality improvement in organisations. It recognises the role that organisational culture plays in data and information management and establishes good practices for improving existing data and information quality infrastructure.

3. ISO Standards

3.1. ISO 8000

The ISO 8000 series of standards was developed to set the global benchmark for data quality and has seen widespread use [53–56]. This standards series issues frameworks for improving data quality specific to different data types, with a focus on the quality of enterprise master data [1]. These are data that are commonly used by organisations to manage information—which differs between organisations—that is critical to their operations. Master data can contain information on products, services, materials, clients, or financial transactions.

The series is organised into several parts that address specific data quality aspects and scenarios. These aspects and scenarios range from general principles of master data quality to specific applications, such as transactional data quality and product data quality. The data quality dimensions included in the ISO 8000 series are detailed in ISO 8000-8 and comprise accuracy, completeness, consistency, timeliness, uniqueness, and validity [57].

ISO 8000 also imports and incorporates notions such as the PDCA (Plan, Do, Check, Act) cycle, as shown here in Figure 2, which was outlined in ISO 9001 to improve data quality [4]. This cycle shares the same fundamentals as the DMAI cycle as described in Section 2.1 for the TDQM framework [58]. The planning stage aims to identify the relevant data quality dimensions for the organisation or task. The implementation stage entails the collection and processing of data. In the checking phase, the data quality dimensions considered in the planning stage are measured on the collected data. Lastly, the acting phase is implemented to continuously improve the processes of the full cycle. The practical application of the PDCA cycle is also described in ISO 8000-61 and provides a robust data quality management framework [59].

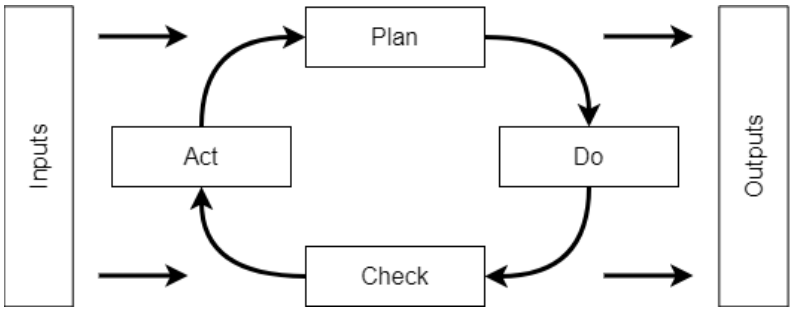


Figure 2. "Plan, Do, Check, Act" cycle, as outlined by the ISO 8000 and ISO 9000 series [4] The cycle follows an iterative process leading to incremental change and improvements and can be applied to design better data management processes.

3.2. ISO 25012

The ISO 25012 standard is a part of the SQuaRE series of International Standards [5,60]. The latter establishes a general-purpose data quality model that can be applied to data stored within a structured computer system [61–66]. ISO 25012 can be specifically used to establish data quality requirements, define quality measures, and plan and perform data quality assessments. It performs a similar role to ISO 8000, but its application is more specific, primarily designed to be compatible with software development applications, processes, and pipelines.

Data quality dimensions in ISO 25012 are classified into fifteen unique characteristics: accuracy, completeness, consistency, credibility, currentness, accessibility, compliance, confidentiality, efficiency, precision, traceability, understandability, availability, portability, and recoverability. These dimensions are placed into a perspective spectrum that ranges from inherent to system-dependent. Inherent data quality dimensions are those that are intrinsic to the data regardless of the context of application and use, while system-dependent dimensions rely on the system and conditions of use to assess the quality of data. Some dimensions fall strictly into one of the two classes, while others need contextual consideration, balancing between inherent and system-dependent characteristics [5].

The comprehensive approach of ISO 25012 enables organisations to employ a single framework for consistently assessing the quality of their data under the assumption that the data are contained within a structured digital system. The dual perspective of inherent and system-dependent data quality dimensions offers a system for identifying the relevant data quality dimensions based on the application or use case while maintaining the sufficient granularity provided by the 15 individual dimensions for data quality assessment without redundancy [5]. However, the general approach of ISO 25012 may not suit organisations with highly specific and unique quality considerations.

4. Government and International Standards

4.1. Fair Information Practice Principles (FIPPS)

The Fair Information Practice Principles (FIPPS) are a set of guidelines established by the Federal Privacy Council (FPC), originally developed in 1973 by the United States Department of Health, Education, and Welfare, to address growing concerns about data privacy and the use of personal data, particularly when data are being used by automated systems [7]. These principles—transparency, individual participation, authority, purpose specification and use, limitation minimisation, access and amendment, quality and integrity, security, and accountability—see widespread use, acting as a foundational framework for ensuring data privacy and protection [67–86]. While not of regulatory status themselves, these principles have significantly influenced privacy legislation and policies, such as the General Data Protection Regulation (GDPR) and the Federal Information Processing

Standards (FIPS) series developed by the National Institute of Standards and Technology (NIST) [87,88]. Notably, several FIPS standards are relevant for preserving data quality while addressing privacy and security concerns:

- FIPS 180-4: Specifies secure hashing algorithms used to generate message digests, which help detect changes in messages and ensure data integrity during transfer and communication [89].
- FIPS 199: Provides a framework for categorising federal information and information systems based on the level of confidentiality, integrity, and availability required. It helps in assessing the impact of data breaches and ensuring appropriate security measures [90].
- FIPS 200: Outlines minimum security requirements for federal information and information systems, promoting consistent and repeatable security practices to protect data integrity and privacy [91].

Organisations wanting to use the FIPPS framework must assess the quality of personally identifiable information (PII) according to the principles it outlines for its effective use. This involves ensuring that data are accurate, relevant, and complete, as well as being collected and used transparently and with proper authority. By adhering to these principles, organisations can maintain high standards of data quality, which is essential for protecting privacy and ensuring the reliability of data-driven decisions.

4.2. Quality Assurance Framework of the European Statistical System (ESS QAF)

The Eurostat Quality Assurance Framework (QAF) is part of the Total Quality Management Framework that specifically addresses the quality of statistically generated outputs and data. It assesses the quality of these data according to five core principles, which are Relevance, Accuracy and Reliability, Timeliness and Punctuality, Coherence and Comparability, and Accessibility and Clarity.

The QAF itself is aligned with the European Statistics Code of Practice [8] and holds regulatory weight [19] under the Treaty on The Functioning European Union [20]. This alignment ensures that the statistical processes, outputs, and data adhere to high standards of quality, integrity, and reliability, which are essential for informed decision making and policy formulation within the European Union.

The QAF, as a data quality framework, focuses primarily on statistical processes and methods. This focus means it has reduced applicability compared to more general frameworks such as TDQM and TQdM, which were discussed previously. However, it shares similarities with these frameworks in its emphasis on key quality dimensions such as accuracy, timeliness, and relevance. This framework has thus been widely used and adapted both inside and outside of the European Union, demonstrating its robustness and flexibility in various contexts [92–105]. The QAF's targeted approach ensures that European statistics are produced according to rigorous standards, making them reliable and useful for decision making and policy formulation within the European Union.

4.3. The UK Government Data Quality Framework

The UK Government Data Quality Framework, published in December 2020, addresses widespread concerns about data quality in the public sector. Motivated by the need to improve decision making, policy formation, and public services, it aims to standardise and enhance data quality practices across government organisations [9]. The framework consists of two parts: conceptual framework and practical guidance. The conceptual framework provides the structure for understanding and approaching data quality. It emphasises five data quality principles, describes the data lifecycle, shown here in Figure 3, and outlines the six core data quality dimensions used to evaluate the quality of data [9,21,106].

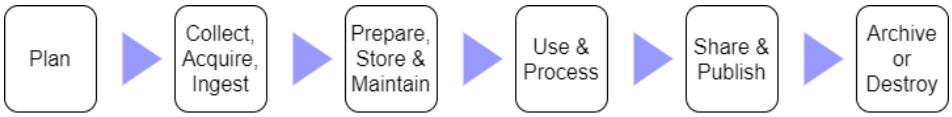


Figure 3. Data lifecycle as outlined by the the UK Government Data Quality Framework. It describes "the different stages the data will go through from design and collection to dissemination and archival/destruction" [9].

The five data quality principles are Commit to data quality, Know your users and their needs, Assess quality throughout the data lifecycle, Communicate data quality clearly and effectively, and Anticipate changes affecting data quality [9,21]. These principles are designed to create accountability and commit to ongoing assessment, improvement, and reporting of data quality. They promote understanding and prioritising user requirements to ensure that data are fit for purpose, focusing on quality measures and assurance at each stage of the data lifecycle. This ensures that end users understand data quality issues and their impact on data use, as well as helps them plan for and prevent future data quality issues through effective change management.

The conceptual framework also outlines a six-stage data lifecycle model consisting of the following activities: plan; collect, acquire, and ingest; prepare, store, and maintain; use and process; share and publish; and archive or destroy [9,21]. This lifecycle model helps identify potential quality issues at each stage and includes guidance on data management practices, quality considerations, and potential problems to address. Lastly, the conceptual framework defines six data quality dimensions as completeness, uniqueness, consistency, timeliness, validity, and accuracy [9,21]. It also provides examples on how to quantify them [106].

The second part of the framework provides practical tools and techniques for implementing the concepts introduced by the conceptual framework [21]. It includes guidance on data quality action planning, root cause analysis, metadata management, communicating data quality to users, and using data maturity models. These tools are designed to help organisations assess, improve, and communicate data quality effectively, supporting the principles and concepts outlined in the conceptual framework. The outputs from the practical guidance are then translated into a maturity assessment which places the data quality infrastructure of the assessed sector or organisation into a scale from 1 (unacceptable) to 5 (optimised) [21,106]. This scale acts as a guideline for improvement of data quality and its supporting infrastructure.

Data Management Body of Knowledge (DAMA DMBoK)

DAMA International is an organisation that specialises in advancing data and information best practices [22,23]. In 2009, it published the Data Management Body of Knowledge (DAMA DMBoK), which comprises a detailed set of guidelines for addressing data management challenges [22]. Among these guidelines, the DMBoK contains a functional framework and a common vocabulary for assessing data quality [107].

While not regulatory by itself, the DMBoK is heavily imparted in other frameworks, such as the UK Government Data Quality Framework. The DMBoK covers multiple data management knowledge areas that are leveraged by the UK Government in their own framework to understand the maturity levels of current data quality and management practices across different sectors. This allows the UK Government to identify areas for improvement and assess the requirements for carrying these improvements [9,108].

The knowledge areas covered by the DMBoK are Data Governance, Data Architecture, Data Modelling and Design, Data Storage and Operations, Data Security, Data Integration and Interoperability, Documents and Content, References and Master Data, Data Warehousing and Business Intelligence, Metadata, and Data Quality. The wide scope covered

by these knowledge areas makes DMBoK a robust and adaptable framework for a diverse range of applications. The five data quality principles used by the UK Government in their own DQF, discussed in Section 4.3, were imported from DMBoK [108].

5. Financial Frameworks

5.1. IMF Data Quality Assessment Framework (DQAF)

The International Monetary Fund (IMF)'s Data Quality Assessment Framework (DQAF), first published in 2003 and updated in 2012, is a comprehensive methodology for assessing data quality specific to the financial sector and institutions. The framework is grounded in the Fundamental Principles of Official Statistics of the United Nations [10] and describes best practices accepted by the international community, such as the use of accepted methodologies, for assessing the quality of data.

The methodologies used by DQAF focus on the quality of statistical systems, processes, and products. As a result, the DQAF is defined by six quality dimensions: quality prerequisites, assurances of integrity, accuracy and reliability, serviceability, and accessibility [10].

Each dimension is further subdivided into elements that can be defined by specific indicators, such as legal and institutional support, adequate resources, relevance to user needs, professionalism, transparency, ethical standards, alignment with international standards, sound statistical techniques, timely and consistent data, and comprehensive metadata. These indicators can be quantified and used to describe the quality of data according to each dimension.

The main aim of the DQAF is to leverage these indicators to enhance the quality of data provided to the IMF by different countries in a standardised way [109]. This promotes financial institution transparency, which supports financial stability and aligns economic policies across participating countries. The DQAF is used and complemented by other IMF data dissemination standards, such as the General Data Dissemination System and the Special Data Dissemination Standard [110], to further advance the goal of using data in promoting transparency and supporting informed decision making.

5.2. Basel Committee on Banking Supervision Standard (BCBS 239)

The Basel Committee on Banking Supervision (BCBS 239)'s Principles for Effective Risk Data Aggregation and Risk Reporting, published in 2013, act as a framework for enhancing risk management practices in banking systems [11]. While BCBS 239 does not explicitly focus on data quality, it incorporates numerous terms and definitions relevant to data quality. This framework addresses the challenges many banks faced in aggregating and reporting risk data effectively during the 2008 financial crisis [111]. BCBS 239 comprises 14 principles organised into five key categories [112]. These categories are "Overarching Governance and Infrastructure"; "Risk Data Aggregation Capabilities"; "Risk Reporting Practices"; "Supervisory Review, Tools, and Cooperation"; and "Implementation Timeline and Transitional Arrangements". The second category, "Risk Data Aggregation Capabilities", focuses on the technical aspects of collecting, processing, and consolidating risk data, and it sets standards for data accuracy, adaptability, clarity, completeness, integrity, and timeliness [113,114].

The focus on these data quality dimensions is due to their relevance for effective risk management [115]. Accuracy reflects the closeness of data to the true values; adaptability refers to the ability to adjust to changing circumstances; clarity ensures that reports and data are easily understood; completeness is aimed at ensuring the availability of relevant data across all organisational units; integrity focuses on safeguarding data from unauthorised changes; and timeliness refers to data availability within the necessary timeframe for

decision making. Together, these elements contribute to a robust framework for enhancing risk data aggregation and reporting practices.

By implementing the principles outlined in BCBS 239, banks are expected to enhance their risk management capabilities, increase transparency, and bolster their resilience to financial shocks [114–116]. This framework is particularly applicable to Global Systematically Important Banks (G-SIBs), which are subject to additional regulatory requirements, and encourages national authorities to extend their principles to Domestic Systematically Important Banks (D-SIBs) as well.

6. Healthcare Frameworks

6.1. ALCOA+ Principles

The ALCOA+ principles—attributable, legitimate, contemporaneous, original, accurate, plus complete, consistent, lasting, and available—establish a comprehensive framework for ensuring data integrity in regulated industries, with particular emphasis on medicine manufacturing [24]. These guidelines outline data quality aspects to be maintained throughout the data lifecycle [117]. The need for such a framework emerged from the growing complexity of data management in an increasingly digital landscape, where the risk of errors has risen in recent years [12,117,118].

The need for ALCOA+ arose from a multifaceted set of challenges in regulated industries [119,120]. Various factors contribute to poor data quality, including human and system errors, inadequate procedures, and inconsistencies across different platforms and processes. ALCOA+ addresses these challenges by offering clear data quality guidelines that enable organisations to implement robust processes, enhancing the overall trustworthiness of their data and data-driven decisions [12,117,118]. Additionally, ALCOA+ helps identify deliberate falsification of medical data, which, if undetected, can lead to severe consequences such as compromised patient safety, inaccurate clinical trial results, or regulatory noncompliance [24].

The U.S. Food and Drug Administration (FDA) has played a significant role in promoting and enforcing the ALCOA+ principles [24]. By adopting these guidelines, the FDA has established a clear standard for evaluating the reliability and trustworthiness of data submitted for regulatory purposes [24]. This adoption has far-reaching implications, as it guides FDA inspections and audits of regulated facilities and serves as a benchmark for compliance with good manufacturing practices (GMPs) and good laboratory practices (GLPs).

6.2. WHO Data Quality Assurance

The Data Quality Assurance (DQA) framework established by the World Health Organisation (WHO) provides a systematic approach for reviewing and enhancing data quality across healthcare facilities worldwide [13]. This framework is designed to identify weaknesses in data management systems and monitor data quality performance through structured processes. It encompasses routine data quality assurance activities, including regular reviews, discrete cross-sectional assessments, and periodic in-depth evaluations tailored to specific programs—all conducted through desk reviews or site assessments [121,122].

Desk reviews focus on analysing the completeness and consistency of existing aggregated data using established WHO metrics, such as completeness, internal consistency, external consistency with other data sources, and alignment with population data [121]. Site assessments evaluate the accuracy of health data through on-site evaluations guided by a checklist that examines data accuracy across reporting hierarchies and assesses the healthcare system's capacity to generate quality data [122]. These assessments are routinely conducted by health facility staff on a monthly basis, with district-level staff performing

periodic evaluations [123]. Both assurance methods use trace indicators as quantifiable measures to assess adherence to data quality standards; meeting established benchmarks indicates that data quality is satisfactory [124–126].

Implementing the DQA framework necessitates a coordinated effort involving all key stakeholders, reflecting the framework's emphasis on data quality as a systemic issue influenced by interactions among various data quality activities. However, the framework does not primarily focus on overarching planning, technical support, funding, or promotional efforts.

7. Discussion

The frameworks presented in this review raise varying considerations for the data quality dimensions they include, depending on their respective domains of application. While differences among these frameworks were anticipated, identifying similarities and gaps in data quality coverage is a crucial aspect of data quality research. The regulatory nature of these frameworks significantly influences how quality data are described, communicated, and used by domain experts. Ultimately, any framework employed—whether developed in-house or adopted from existing frameworks—must align with the regulations set for their specific domains of application.

To facilitate a one-to-one comparison across all examined frameworks, standardised data terminology is needed. We employed a previously developed standardised data quality framework to achieve this comparison [2], which is found in Table 1. This approach uses the definitions of each individual data quality dimension, as outlined by their respective frameworks, and maps them to the corresponding definition in our framework to find the common name for that dimension. This standardisation is necessary because different frameworks describe the same data quality dimension using different terminology. Additionally, some dimensions, as specified by each framework, cover different aspects of the same overarching concept. For example,"enduring" (ALCOA+) [11,111–114] and "distribution" (BCBS-239) [12,24,117,118] both relate to governance but address different aspects of it. This highlights the importance of a standardised language to ensure clarity and usability across diverse frameworks.

Table 1. Mapping data quality dimensions across frameworks: The first column contains the labels for the DQFs reviewed in this work. The second column lists only those dimensions explicitly described by the frameworks. The third column provides a standardised DQ nomenclature, as detailed by Miller et al. [2], for the data quality dimensions mentioned by the frameworks. Grey shading helps visualise instances where the mapping between the second and third column terms is not one-to-one.

Framework	DQ Dimensions	DQ Common Nomenclature
TDQM	Accuracy	Accuracy
	Objectivity	Precision
	Believability	Credibility
	Reputation	Credibility
	Access	Accessibility
	Security	Confidentiality
	Relevancy Value-added	Usefulness
	Timeliness	Currenctness
	Completeness	Completeness
	Amount of Data	Quantity
	Interpretability Ease of Understanding	Understandability
	Concise Representation	Precision
	Consistent Representation	Consistency

Table 1. *Cont.*

Framework	DQ Dimensions	DQ Common Nomenclature
ISO 8000	Accuracy	Accuracy
	Completeness	Completeness
	Consistency	Consistency
	Timeliness	Currentness
	Uniqueness	Usefulness
	Validity	Credibility
ISO 25012	Accuracy	Accuracy
	Accessibility	Accessibility
	Availability	Availability
	Completeness	Completeness
	Compliance	Compliance
	Confidentiality	Confidentiality
	Consistency	Consistency
	Credibility	Credibility
	Currentness	Currentness
	Efficiency	Efficiency
	Precision	Precision
	Portability	Portability
	Recoverability	Recoverability
	Traceability	Traceability
	Understandability	Understandability
FIPPS	Accuracy	Accuracy
	Relevancy	Usefulness
	Timeliness	Currency
	Completeness	Completeness
ESS QAF	Statistical Confidentiality and Data Protection	Confidentiality
	Accessibility and Clarity	Accessibility
		Availability
		Understandability
		Traceability
	Relevance	Usefulness
	Timeliness and Punctuality	Currentness
	Accuracy and Reliability	Accuracy
	Impartiality and Objectivity	Credibility
	Cost Effectiveness	Efficiency
	Coherence and Comparability	Consistency
UK GOV DQF (DAMA DMBoK)	Completeness	Completeness
	Consistency	Consistency
	Timeliness	Currentness
	Uniqueness	Usefulness
	Validity	Credibility
	Accuracy	Accuracy
IMF	Prerequisites of quality	Usefulness
	Assurance of Integrity	Credibility
		Traceability
	Methodological Soundness	Semantics
	Accuracy and Reliability	Accuracy
	Serviceability	Consistency
		Currency
		Traceability
	Accessibility	Accessibility
		Understandability
BCBS 239	Accuracy	Accuracy
	Clarity and Usefulness	Understandability
	Comprehensiveness	Completeness
	Frequency	Currentness
	Distribution	Governance

Table 1. *Cont.*

Framework	DQ Dimensions	DQ Common Nomenclature
ALCOA+	Accurate	Accuracy
	Attributable	Traceability
	Available	Availability
	Complete	Completeness
	Consistent	Consistency
	Enduring	Governance
	Legible	Understandability
	Original	Traceability
WHO	Completeness	Completeness
	Timeliness	Currentness
	Internal Consistency	Consistency Accuracy
	External Consistency	Consistency Accuracy
	Consistency of Population Data	Governance

Figure 4 offers a condensed view of the information provided by Table 1, which was made possible by the mapping exercise using our previously proposed data quality framework [2]. This figure presents a matrix grid that visually represents the presence or absence of data quality dimensions across the various frameworks reviewed in this study. The vertical axis has the data quality frameworks—in the same order they were introduced in this paper—while the horizontal axis contains all data quality dimensions considered. Blue cells indicate that a particular data quality dimension is included in the framework, while red cells indicate that it is not considered.

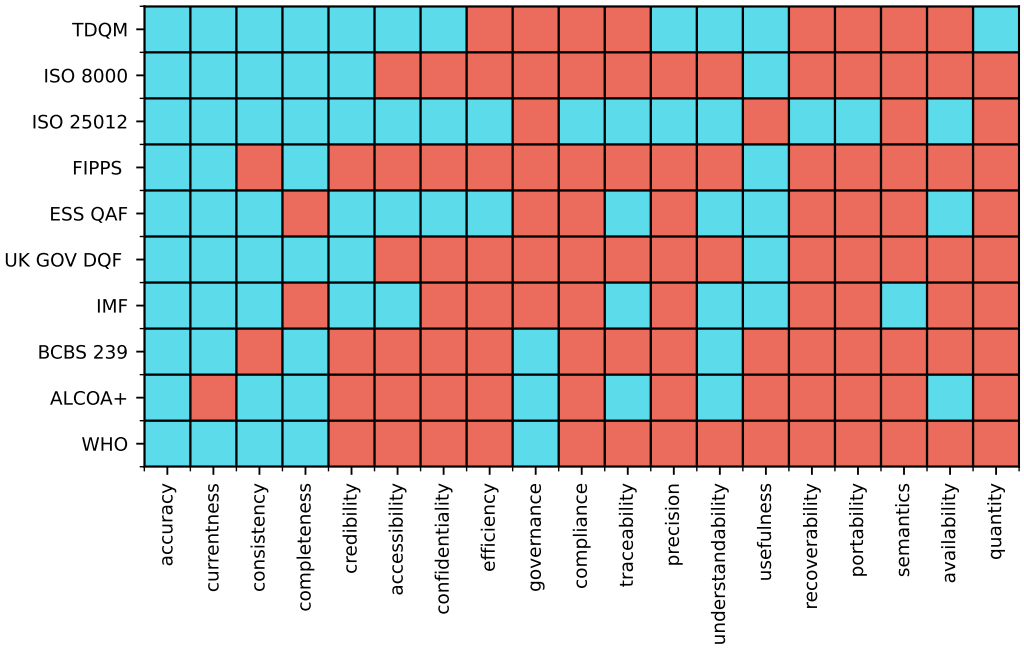


Figure 4. Gaps in data quality dimensions of the reviewed frameworks. Data quality dimensions present in specific frameworks are highlighted in blue, while absent dimensions are indicated in red.

From Figure 4, it is noticeable that frameworks such as TDQM [6] and ISO 25012 [5] cover a broader range of dimensions—11 and 15 out of a total of 19, respectively—while other frameworks, such as FIPPS [7] and WHO's DQA [13]—4 and 5 out of a total of 19, respectively—have notable gaps in their coverage. This shows a trend where frameworks designed for general or all-purpose applications tend to cover a larger range of data quality dimensions, while frameworks tailored for use in specific domains cover fewer dimensions.

This observation is based in the sample size comprised of the frameworks reviewed, so it should be used with caution. However, the trend suggests that while general-purpose frameworks may offer broader coverage, specialised frameworks can provide targeted insights that are crucial for specific applications. While at first it can seem to indicate that specialised frameworks are more limited than broader frameworks, it is important to note that this can also make their use more streamlined, as they cover data quality dimensions more relevant for their specific domains of application.

Another relevant aspect to consider is that more specific frameworks often incorporate data quality dimensions that are not present in all-purpose frameworks. The IMF's DQAF, BCBS 239, ALCOA+, and WHO's DQA frameworks cover "governance", "usefulness", and "semantics" as dimensions of data quality [10,11,13,24,111]. All of these dimensions are absent from ISO 25012 [5], with the most comprehensive DQF reviewed in work using the number of data quality dimensions as a metric. The inclusion of these dimensions in frameworks such as ALCOA+ and WHO's DQA demonstrates the need to address specific data quality concerns that can be overlooked in more generalisable frameworks. Additionally, the data quality concerns addressed by the inclusion of these dimensions in specialised frameworks are not consistent across frameworks. We previously discussed how "enduring" (ALCOA+) and "distribution" (BCBS-239) both relate to "governance" but address different facets pertinent to their respective domains: these being healthcare and finance [11,24,111]. While there is a risk of losing granularity in assessing data quality, aggregating data quality dimensions under a common terminology facilitates a comparison between frameworks that would otherwise not be feasible [2]. Notwithstanding this limitation, the presence of less common data quality dimensions in specialised frameworks highlights a potential understated value of these frameworks, which is that they can showcase the need for data quality dimensions that otherwise might not be included in all-purpose frameworks [2].

The only framework that accounted for "quantity" as a dimension of data quality was TDQM. This framework is the oldest covered in this review and one of the earliest examples of a structured DQF found [6]. The capacity of data storage systems has increased with time [127]; hence, the "quantity" of data played a larger role in overall data quality in the earlier days of computing, which can help explain its absence from more modern frameworks. However, with the growth of big data applications, data generation has been regularly outpacing data storage capacity [128]. As organisations increasingly deal with big data challenges, the relevance of "quantity" as a dimension of data quality is becoming more noticeable. This is exemplified by the increased generation of healthcare data, pushing forward the need to assess the quality of data in relation to available storage solutions and mechanisms [129,130].

One last noteworthy aspect is that core data quality dimensions, such as "accuracy", "completeness", "consistency", and "timeliness" have a greater representation across all reviewed frameworks. Their prevalence reflects their historical weight, as these dimensions have been imported from older frameworks into modern ones. However, it is important to note that while these core dimensions are consistently represented, their definitions and metrics have evolved over time and across different domains. For instance, the definition of accuracy in early frameworks might have focused on the correctness of data entries, whereas modern interpretations could include aspects of data precision and reliability in complex datasets. Similarly, timeliness has evolved from simply ensuring data are up-to-date to encompassing real-time data processing and availability in dynamic environments [1]. Despite this, the rapid emergence of new technologies highlights the need for newer dimensions to be recognised and integrated into all-purpose data quality frameworks.

This is exemplified by the dimension of "semantics", as semantic technologies, such as knowledge graphs and ontologies, have become more relevant to big data applications [131]. For instance, in healthcare, semantic frameworks can be used to design knowledge systems that are interoperable, allowing multiple stakeholders collaborating on the same processes to understand and validate each other [132]. This interoperability is crucial for ensuring that data are accurately interpreted and used across different systems and organisations. Another relevant example of the use of "semantics" comes from the finance sector, where tools such as ontologies and graph databases can support frameworks for bankruptcy prediction [133]. These applications require data to meet pre-established criteria such as detailed descriptions and nomenclature, further highlighting the need for a way to evaluate the quality of these data.

The rapid growth of AI technologies such as LLMs provides an example of why these less-represented dimensions are crucial. AI models rely on high-quality data to function effectively. Dimensions such as "quantity", "availability", "semantics", "portability", and "compliance" are very relevant in the AI domain. For instance, LLMs require large quantities of available and accessible data to learn from [134], and having data semantically linked is required for allowing these models to understand and process natural language [135,136]. Having data with a high degree of portability allows these data to be used across different systems and platforms, while "compliance" plays an inherent role in defining data quality specifically for AI systems, as it ensures that data meet legal and ethical standards [137].

This analysis of regulation-backed data quality frameworks reveals both strengths and limitations to their design and application. While core dimensions remain foundational across various frameworks, the emergence of new technologies imposes the inclusion of additional dimensions such as "semantics" to address the evolving landscape of data management and quality. The insights gained from this review highlight the importance of adopting a standardised approach to data quality that accommodates both established and emerging dimensions. By doing so, organisations can enhance their data management practices, ensure compliance with regulatory standards, and ultimately improve decision-making processes in an increasingly data-driven world.

8. Conclusions

The goal of this work was to review and compare different data quality frameworks that are underpinned by regulatory, statutory, or governmental standards and regulations. By examining frameworks such as TDQM, ISO 8000, ISO 25012, and others, this work sought to identify how these frameworks define, measure, and apply data quality dimensions. The focus on regulation-backed frameworks also allowed for the identification of frameworks used in heavily regulated industries, such as the IMF's DQAF and BCBS 239 for the financial sector and WHO's DQA and ALCOA+ for the healthcare sector. Understanding how data quality is applied across a varied landscape of domains, including their regulatory requirements, promotes the development of tools for applying these frameworks that comply with current requirements set by standards and regulations. Knowledge of how data quality is leveraged across specific domains is also valuable for informing regulators of emerging technologies such as AI systems [3].

A standardised data quality model, which was first suggested by the authors [2], was used to connect all the data quality dimensions found in each framework to a common vocabulary. This was done by looking at the different definitions that each framework gave to its dimensions and aligning them with dimensions sharing identical or similar definitions from other frameworks. The outcomes of this approach are shown in a matrix

grid in Figure 4, with frameworks and dimensions on the vertical and horizontal axes, respectively. This made it possible to compare frameworks using a gap analysis.

Findings from the bibliometric analysis of all reviewed frameworks reveal that core data quality dimensions, such as "accuracy", "completeness", "consistency", and "timeliness", are well represented across all frameworks. Other dimensions, such as "semantics" and "quantity", are overlooked by most frameworks; however, they highlight the need for modern dimensions to be considered. "Semantics" is a direct response to the increased use of knowledge graphs and ontologies, while "quantity" correlates with the capacity of data storage systems being outpaced by data generation. Additionally, frameworks tailored for specific domains, such as finance and healthcare, were found to often include data quality dimensions that were absent from more general frameworks due to specific industry needs. This highlights the importance of adopting a standardised approach to data quality that takes into account both established and emerging data quality dimensions and promotes data governance and compliance in data-rich environments.

This review provides insight into the data quality frameworks currently employed across a varied landscape of domains, including highly regulated industries. It fills the gaps of other modern reviews of this subject area by providing a like-for-like comparison of data quality frameworks used in regulated industries, without limiting the scope of these frameworks to framework types (all-purpose frameworks are reviewed alongside more specific frameworks) or to domains of applications (frameworks from multiple industries are reviewed). This study also highlights the need to develop and integrate more modern dimensions of data quality in existing frameworks to keep up with the needs of emerging technologies, such as is the case for AI systems based on LLM models. It also offers guidance for the creation of tools and applications that use these frameworks by highlighting the data quality dimensions that specific industries and domains have as set requirements. Lastly, this review also highlights the need to consider the inclusion of emerging data dimensions to enable established all-purpose frameworks to keep up with the rapidly evolving technological landscape.

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